Important resources for Quantum Machine Learning:

1. if you need some ML refreshment then please follow this link:
    a. https://jakevdp.github.io/PythonDataScienceHandbook/05.00-machine-learning.html. This free online book has a great introduction to python and data science in general. Besides, it contains great material on Scikit-Learn.
2. If you need a good introduction to Linear Algebra and the mathematics behind ML:
    a. https://www.coursera.org/specializations/mathematics-machine-learning. This specialization will show the most important topics of which you should be aware.
3. If you need a good introduction to Quantum Computing:
    a. Introduction to quantum computing in Arabic
    b. Alexandria Quantum Computing Group
    c. https://qiskit.org/textbook/preface.html. Qiskit textbook is a great place to start your journey to the QC realm. It's intuitive, straightforward, and interactive.
    d. https://www.youtube.com/playlist?list=PLnhoxwUZN7-6hB2iWNhLrakuODLaxPTOG. Quantum Mechanics and Quantum Computation course by Prof Umesh Vazirani.
    e. Quantum Computing Explained. Link. this is a great book to start with many solved examples.
4. Quantum Machine Learning course by Prof Peter Wittek "May God rest his soul":
    a. https://www.youtube.com/playlist?list=PLmRxgFnCIhaMgvot-Xuym_hn69lmzIokg
5. Qiskit Tutorial about combining PyTorch with Qiskit framework by Amira Abbas who is an active member of Qiskit Advocate team:
    a. https://qiskit.org/textbook/ch-machine-learning/machine-learning-qiskit-pytorch.html
6. How to calculate Quantum Gradients and use them for classification:
    a. https://bit.ly/2CtVVkO
7. How to use Qiskit in an optimization problem "Solving IBM Quantum Challenge by machine learning":
    a. https://bit.ly/2ZHksvN
8. This is an external resource but it's very rich in its content:
    a. https://pennylane.ai/qml/demonstrations.html. PennyLane is a QML framework that makes it easy to calculate gradients. It can also be integrated easily with Tensorflow and PyTorch frameworks. Another important feature is that it can be integrated with Qiskit. Just imagine the possibilities.
    b. Another useful tutorial on Hybrid architectures: https://bit.ly/30oiHmm
9. Variational quantum classifiers (VQC): https://ibm-q4ai.mybluemix.net/, https://arxiv.org/abs/1804.00633
10. quantum natural language processing
    a. https://www.youtube.com/watch?v=g6P1MprxS2I
    b. https://www.youtube.com/watch?v=mL-hWbwVphk
    c. https://www.youtube.com/watch?v=6-uKZN5_eAU
    d. https://arxiv.org/pdf/2005.04147.pdf

    e. https://medium.com/cambridge-quantum-computing/quantum-natural-language-processing-748d6f27b31d

11. You should spend some time on the documentation https://qiskit.org/documentation/ where you will find great updated tutorials for different topics.
12. Great online sessions:
    a. https://www.youtube.com/watch?v=dtLjvGqPoVM&t=4s
    b. https://www.youtube.com/watch?v=sS5ovtbXDGQ&t=4s
    c. https://www.youtube.com/watch?v=Lbndu5EIWvI
    d. https://youtu.be/pe1d0RyCNxY
    e. https://youtu.be/Ycy40s4aXxs
    f. https://www.youtube.com/watch?v=Xh9pUu3-WxM
    g. https://youtu.be/_M2GQAknykg
    h. https://youtu.be/-DWng3jyBIM
    i. https://youtu.be/M1aKwb6bHW4